Characterization of transcriptional responses to environmental stress by differential location analysis

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ABSTRACT

Unicellular organisms like yeast, need to rapidly respond to environmental condition changes for their survival. Using high-throughput location analysis (Chromatin Immuno-Precipitation on DNA chip, or ChIP-Chip in short), Harbison *et al.*[1] determined the genomic binding locations of 204 transcription factors (TFs) from the yeast *Saccharomyces cerevisiae* in rich media condition and 12 stress conditions. Here, we report a statistical method for *differential location analysis*, to determine the set of regulators that bind to significantly different genomic regions under certain stress conditions. From the published ChIP-Chip data by Harbison *et al.*, we were able to identify 105 TFs-condition pairs which showed statistically significant differential binding patterns ($p < 0.05$). Comparison with published Microarray data revealed that the expression levels of nearly half of these TFs did not significantly change under the corresponding environmental stress, which implies that such regulatory responses would not be revealed solely by Microarray data. In conclusion, complementary differential analyses (e.g. differential location analysis) are required, in addition to commonly used Microarray-based differential expression analysis, in order to illustrate the global picture of cellular responses to environmental stresses.

DATA

ChIP-on-Chip : 204 Yeast Transcription Factors (TFs)[1]
1 rich medium condition + **12** stress conditions
6540 genes

Microarray

Stress Condition	Concentration	Time point	Array	Ref.
H2O2 – oxidation	0.3 mM	20 mins	cDNA	2
H2O2 – oxidation	0.32 mM	20 mins	cDNA	3
H2O2 - oxidation	0.4 mM	20 mins	oligo	4
SM (Sulfometuron methyl) – AA starvation	0.2 μg/ml	15 mins	cDNA	5
SM (Sulfometuron methyl) – AA starvation	0.2 μg/ml	240 mins	cDNA	5

Differential Binding

Differential binding is represented as P_i^k by using ChIP-chip binding ratio data between stress condition (A) and rich medium (B).

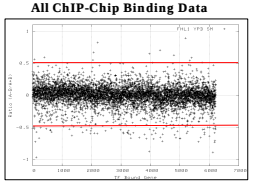
$$P_i^k = \frac{A_i - B_i}{A_i + B_i}$$

A_i, B_i: binding ratio of TF k to regulated region i under stress and rich medium culture condition respectively

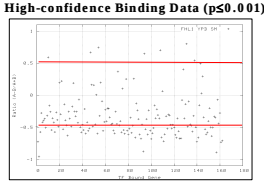
Distribution of P_i^k

Most of P_i^k were within ± 0.5 range mostly around zero. But the high-confidence binding regions (at least one of the two conditions with high confidence p-value ≤ 0.001) mostly showed skewed-distribution that have been further confirmed by Chi-Square Test.

All ChIP-Chip Binding Data



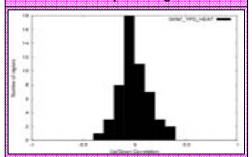
High-confidence Binding Data (p≤0.001)



Chi-Square Test

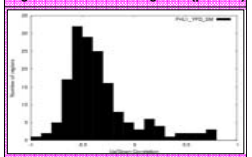
Chi-Square test was performed to check the distribution against normal distribution. P-value ≤ 0.05 is considered to skewed from normal distribution (mean=0).

All ChIP-Chip Binding Data



Normal Dist. (p=0.97)

High-confidence Binding Data (p≤0.001)



Skewed Dist. (p=0.00003)

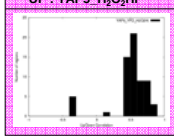
Classification of TFs

Depending on the skewedness of high-confidence binding data, 105 TF-Condition pairs with skewed distribution have been grouped into three categories.

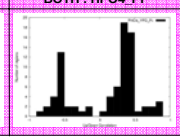
UP : More than 2/3 are in '**+**' direction
DOWN : More than 2/3 are in '**–**' direction
BOTH : Similar proportions of '**+**' and '**-**'

CLASS	UP	BOTH	DOWN
Ratio	Ra > 0.67	0 < Ra or Rb < 0.67	Rb > 0.76
TF & Condition	DIG1_ASN4 MBP1_H2O2Lo HAP4_SM GAT1_HEAT DAL82_SM YAP1_H2O2Lo CBF1_SM AFT2_H2O2Lo CHA4_SM DIG1_BUT14 HAP4_SM FHL1_H2O2Hi RPN4_H2O2Lo GCN4_SM MSN2_H2O2Lo UME6_H2O2Hi MSN2_HEAT YAP1_H2O2Hi HAP4_H2O2Lo GZF3_H2O2Hi GCN4_RAPA CIN5_H2O3Lo SIG1_H2O2Hi PHO1_BUT14 PHO2_Pi YAP6_H2O2Lo SFP1_H2O2Hi GLN3_SM RAP1_SM RDS1_H2O2Hi PHO1_BUT90 ARR1_H2O2Hi YAP6_H2O2Hi MSN4_Acid HEF1_H2O2Hi FHL1_RAPA MSS11_BUT14 RLM1_BUT14 CAD1_SM MBP1_H2O2Hi HAP5_SM NRG1_H2O2Hi MSN2_H2O2Hi THI2_Thi- GZF3_RAPA STE12_BUT90 GAL4_RAFF RDS1_H2O2Hi PHO2_SM RTG1_SM STE12_Alpha GLN3_RAPA DIG1_BUT90 RPH1_H2O2Lo PHO2_H2O2Hi MSN4_H2O2Hi AFT2_H2O2Hi YAP7_H2O2Hi ASH1_BUT14 YAP5_H2O2Hi MET28_SM ROX1_H2O2Hi PUT3_SM STE12_BUT14 YAP7_H2O2Lo DAL82_RAPA YAP1_H2O2Hi BAS1_SM RTG3_SM MET32_SM SOK2_BUT14 REB1_H2O2Lo SKN7_H2O2Hi YAP1_HEAT RIM101_H2O2Lo IME4_H2O2Hi HAP2_RAPA XBP1_H2O2Lo HEF1_H2O2Lo FKH2_H2O2Hi FKH2_H2O2Lo YJL206C_H2O2Hi CIN5_H2O2Hi ARG80_SM MIG2_H2O2Hi SKN7_H2O2Lo MSN2_RAPA MGA1_H2O2Hi MAC1_H2O2Hi ADR1_HEAT REB1_H2O2Hi ROX1_H2O2Lo GAT1_RAPA SFP1_H2O2Hi PDR1_H2O2Lo XBP1_BUT90 RTG3_H2O2Hi UME1_H2O2Hi MAL33_H2O2Hi DAL82_SM STP1_SM MOT3_SM	
Num	28	43	34

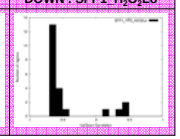
UP : YAP5_H_2O_2Hi



BOTH : HPO4_Pi

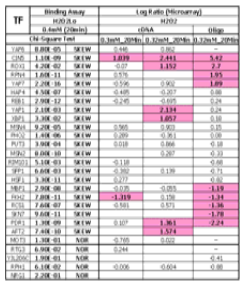


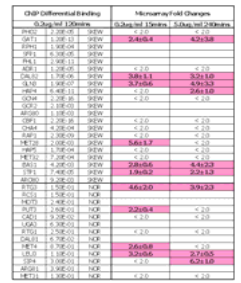
DOWN : SFP1_H_2O_2Lo



Comparison with Microarray

ChIP-Chip and corresponding Microarray data were compared to see the expression levels of transcription factors (TFs). H2O2 Low (oxidation) and SM (Amino acid starvation) stress conditions were used.





ChIP-Chip	H2O2 Lo (27 TFs)								SM (34 TFs)			
	cDNA		oligo		Both		Any		15 Min		4 Hrs	
Microarray	Match	NoMatch	Match	NoMatch	Match		Match		Match	NoMatch	Match	NoMatch
	24	3	22	5	14		28		23	11	23	11
Skewed	21	1	20	2	13		22		15	6	15	6
Fold change >=2	7		9		4	31%	12	55%	6		6	
Fold change 0 ~ 2	14		11		9	69%	10	45%	9		9	
Normal	3	2	2	3	1		4		8	5	8	5
Fold change >=2	0		0						4		3	
Fold change 0~2	3		2		1		4		4		5	

CONCLUSIONS

105 Yeast Transcription Factor (TF) – Stress condition pairs have been found to have differential binding patterns ($p<0.05$) and further grouped into three categories based on skewedness; UP, DOWN, and BOTH. Comparison with Microarray expression have revealed that the gene expression level of almost 50% of TFs do not show any significant change in response to environmental change, which implies the importance of differential binding (not expression of TF). Detailed mechanisms need to be further investigated.

ACKNOWLEDGEMENTS

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REFERENCES

1. Harbison, C.T., et al., *Transcriptional regulatory code of a eukaryotic genome.* Nature, 2004. **431**(7004): p. 99-104. .
2. Shapira, M., E. Segal, and D. Botstein, *Disruption of yeast forkhead-associated cell cycle transcription by oxidative stress.* Mol Biol Cell, 2004. **15**(12): p. 5659-69.
3. Gasch, A.P., et al., *Genomic expression programs in the response of yeast cells to environmental changes.* Mol Biol Cell, 2000. **11**(12): p. 4241-57.
4. Causton, H.C., et al., *Remodeling of yeast genome expression in response to environmental changes.* Mol Biol Cell, 2001. **12**(2): p. 323-37.
5. Jia, M.H., et al., *Global expression profiling of yeast treated with an inhibitor of amino acid biosynthesis, sulfometuron methyl.* Physiol Genomics, 2000. **3**(2): p. 83-92.